5x10 SIMULATION

WITH MOVW, ADDW, AND OUTPUTW

Testbench Module

```
module CU_TB;
        reg PROG_RUN, NEXT, CS, OE, WE, CLK;
        reg [11:0] INST;
        wire [7:0] PORT;
        wire [3:0] PC_CHECK;

        CU U1(PORT,PC_CHECK,INST,PROG_RUN,NEXT,CS,OE,WE,CLK);

        always #1 CLK = ~CLK;

        initial begin
                CLK = 0;
                PROG_RUN = 1; NEXT = 0;
                CS = 0; OE = 0; WE = 1;

                $display("CPU SIMULATION");
                $display("Time\tPC\tINST\t\tW\t\tPORT");
                $display("_____________________________________________");
                $monitor("%0t\t%b\t%b\t\t%b\t\t%b",
                        $time, U1.PC, INST, U1.W, PORT);


                $display("PROGRAM MODE");
```

Cont.

```
                //5X10 THROUGH ADDITION

                //MOV 5 to W            PC 0000
                #2 INST = 12'b001000000101; #1 WE = 0; #1 WE = 1; #2 NEXT = 1; #2; NEXT = 0;

                //ADD 5 to W            PC 0001
                #2 INST = 12'b100100000101; #1 WE = 0; #1 WE = 1; #2; NEXT = 1; #2; NEXT = 0;

                //ADD 5 to W            PC 0010
                #2 INST = 12'b100100000101; #1 WE = 0; #1 WE = 1; #2; NEXT = 1; #2; NEXT = 0;

                //ADD 5 to W            PC 0011
                #2 INST = 12'b100100000101; #1 WE = 0; #1 WE = 1; #2; NEXT = 1; #2; NEXT = 0;

                //ADD 5 to W            PC 0100
                #2 INST = 12'b100100000101; #1 WE = 0; #1 WE = 1; #2; NEXT = 1; #2; NEXT = 0;

                //ADD 5 to W            PC 0101
                #2 INST = 12'b100100000101; #1 WE = 0; #1 WE = 1; #2; NEXT = 1; #2; NEXT = 0;

                //ADD 5 to W            PC 0110
                #2 INST = 12'b100100000101; #1 WE = 0; #1 WE = 1; #2; NEXT = 1; #2; NEXT = 0;

                //ADD 5 to W            PC 0111
                #2 INST = 12'b100100000101; #1 WE = 0; #1 WE = 1; #2; NEXT = 1; #2; NEXT = 0;

                //ADD 5 to W            PC 1000
                #2 INST = 12'b100100000101; #1 WE = 0; #1 WE = 1; #2; NEXT = 1; #2; NEXT = 0;

                //ADD 5 to W            PC 1001
                #2 INST = 12'b100100000101; #1 WE = 0; #1 WE = 1; #2; NEXT = 1; #2; NEXT = 0;

                //OUTPUT W              PC 1010
                #2 INST = 12'b100000000000; #1 WE = 0; #1 WE = 1; #2 NEXT = 1; #2; NEXT = 0;

                #2 PROG_RUN = 0;

                $display("RUN MODE");

                #500 $finish;
        end
endmodule
```

Transcript

```
# CPU SIMULATION
# Time   PC      INST            W               PORT
# ____________________________________________________________
# PROGRAM MODE
# 0      0000    xxxxxxxxxxxx    00000000        00000000
# 1      0000    xxxxxxxxxxxx    00000000        zzzzzzzz
# 2      0000    001000000101    00000000        zzzzzzzz
# 3      0000    001000000101    00000000        xxxxxxxx
# 5      0000    001000000101    00000000        zzzzzzzz
# 7      0001    001000000101    00000000        00000101
# 10     0001    100100000101    00000000        00000101
# 11     0001    100100000101    00000000        xxxxxxxx
# 13     0001    100100000101    00000000        zzzzzzzz
# 15     0010    100100000101    00000000        00000101
# 19     0010    100100000101    00000000        xxxxxxxx
# 21     0010    100100000101    00000000        zzzzzzzz
# 23     0011    100100000101    00000000        00000101
# 27     0011    100100000101    00000000        xxxxxxxx
# 29     0011    100100000101    00000000        zzzzzzzz
# 31     0100    100100000101    00000000        00000101
# 35     0100    100100000101    00000000        xxxxxxxx
# 37     0100    100100000101    00000000        zzzzzzzz
# 39     0101    100100000101    00000000        00000101
# 43     0101    100100000101    00000000        xxxxxxxx
# 45     0101    100100000101    00000000        zzzzzzzz
# 47     0110    100100000101    00000000        00000101
# 51     0110    100100000101    00000000        xxxxxxxx
# 53     0110    100100000101    00000000        zzzzzzzz
# 55     0111    100100000101    00000000        00000101
# 59     0111    100100000101    00000000        xxxxxxxx
# 61     0111    100100000101    00000000        zzzzzzzz
# 63     1000    100100000101    00000000        00000101
# 67     1000    100100000101    00000000        xxxxxxxx
# 69     1000    100100000101    00000000        zzzzzzzz
# 71     1001    100100000101    00000000        00000101
# 75     1001    100100000101    00000000        xxxxxxxx
# 77     1001    100100000101    00000000        zzzzzzzz
# 79     1010    100100000101    00000000        00000101
# 82     1010    100000000000    00000000        00000101
# 83     1010    100000000000    00000000        xxxxxxxx
# 85     1010    100000000000    00000000        zzzzzzzz
# 87     1011    100000000000    00000000        00000000
```

Cont.

```
# RUN MODE
# 91     0000    100000000000    00000000        00000000
# 97     0000    100000000000    00000101        00000000
# 99     0001    100000000000    00000101        00000000
# 107    0001    100000000000    00001010        00000000
# 109    0010    100000000000    00001010        00000000
# 117    0010    100000000000    00001111        00000000
# 119    0011    100000000000    00001111        00000000
# 127    0011    100000000000    00010100        00000000
# 129    0100    100000000000    00010100        00000000
# 137    0100    100000000000    00011001        00000000
# 139    0101    100000000000    00011001        00000000
# 147    0101    100000000000    00011110        00000000
# 149    0110    100000000000    00011110        00000000
# 157    0110    100000000000    00100011        00000000
# 159    0111    100000000000    00100011        00000000
# 167    0111    100000000000    00101000        00000000
# 169    1000    100000000000    00101000        00000000
# 177    1000    100000000000    00101101        00000000
# 179    1001    100000000000    00101101        00000000
# 187    1001    100000000000    00110010        00000000
# 189    1010    100000000000    00110010        00000000
# 193    1010    100000000000    00110010        00110010
# 195    1011    100000000000    00110010        00110010
# 201    1100    100000000000    00110010        00110010
# 207    1101    100000000000    00110010        00110010
# 213    1110    100000000000    00110010        00110010
# 219    1111    100000000000    00110010        00110010
# ** Note: $finish    : C:/intelFPGA_lite/20.1/CPU/CPU_New
#    Time: 590 ps  Iteration: 0  Instance: /CU_TB_New
```

TB1 SIMULATION

WITH MOVW, NOOP, ADDW, JUMP, SUBW, CMP, SKIPE, MOVWM, MOVMW AND OUTPUTW

Testbench Module

```
module CU_TB;
        reg PROG_RUN, NEXT, CS, OE, WE, CLK;
        reg [11:0] INST;
        wire [7:0] PORT;
        wire [3:0] PC_CHECK;

        CU U1(PORT,PC_CHECK,INST,PROG_RUN,NEXT,CS,OE,WE,CLK);

        always #1 CLK = ~CLK;

        initial begin
                CLK = 0;
                PROG_RUN = 1; NEXT = 0;
                CS = 0; OE = 0; WE = 1;

                $display("CPU SIMULATION");
                $display("Time\tPC\tINST\t\tW\t\tPORT");
                $display("_____________________________________________________");
                $monitor("%0t\t%b\t%b\t\t%b\t\t%b",
                        $time, U1.PC, INST, U1.W, PORT);


                $display("PROGRAM MODE");
```

```
                //MOV NOOP ADDW JUMP SUBW CMP SKIPE MOVWW MOVMW OUTPUTW

                //MOV 1 to W            PC 0000
                #2 INST = 12'b001000000001; #1 WE = 0; #1 WE = 1; #2 NEXT = 1; #2; NEXT = 0;

                //NOOP                  PC 0001
                #2 INST = 12'b000000000001; #1 WE = 0; #1 WE = 1; #2 NEXT = 1; #2; NEXT = 0;

                //ADD 3 to W            PC 0010
                #2 INST = 12'b100100000011; #1 WE = 0; #1 WE = 1; #2; NEXT = 1; #2; NEXT = 0;

                //JUMP to PC 0101       PC 0011
                #2 INST = 12'b011100000101; #1 WE = 0; #1 WE = 1; #2 NEXT = 1; #2; NEXT = 0;

                //ADD 1 to W            PC 0100
                #2 INST = 12'b100100000001; #1 WE = 0; #1 WE = 1; #2 NEXT = 1; #2; NEXT = 0;

                //ADD 2 to W            PC 0101
                #2 INST = 12'b100100000010; #1 WE = 0; #1 WE = 1; #2 NEXT = 1; #2; NEXT = 0;

                //SUB 1 from W          PC 0110
                #2 INST = 12'b101100000001; #1 WE = 0; #1 WE = 1; #2 NEXT = 1; #2; NEXT = 0;

                //CMP 5 WITH 5          PC 0111
                #2 INST = 12'b010100000101; #1 WE = 0; #1 WE = 1; #2 NEXT = 1; #2; NEXT = 0;
```

Cont.

```
                //SKIPE -- SHOULD SKIP  PC 1000
                #2 INST = 12'b011000000000; #1 WE = 0; #1 WE = 1; #2 NEXT = 1; #2; NEXT = 0;

                //ADD 1 to W            PC 1001
                #2 INST = 12'b100100000001; #1 WE = 0; #1 WE = 1; #2 NEXT = 1; #2; NEXT = 0;

                //MOV W to M            PC 1010
                #2 INST = 12'b001100000111; #1 WE = 0; #1 WE = 1; #2 NEXT = 1; #2; NEXT = 0;

                // ADD 2 to W           PC 1011
                #2 INST = 12'b100100000010; #1 WE = 0; #1 WE = 1; #2 NEXT = 1; #2; NEXT = 0;

                //MOV M to W            PC 1100
                #2 INST = 12'b010000000111; #1 WE = 0; #1 WE = 1; #2 NEXT = 1; #2; NEXT = 0;

                //OUTPUT W              PC 1101
                #2 INST = 12'b100000000000; #1 WE = 0; #1 WE = 1; #2 NEXT = 1; #2; NEXT = 0;

                #2 PROG_RUN = 0;

                $display("RUN MODE");

                #500 $finish;
        end
endmodule
```

Transcript

```
# CPU SIMULATION
# Time   PC      INST            W               PORT
# ____________________________________________________________
# PROGRAM MODE
# 0      0000    xxxxxxxxxxxx    00000000        00000000
# 1      0000    xxxxxxxxxxxx    00000000        zzzzzzzz
# 2      0000    001000000001    00000000        zzzzzzzz
# 3      0000    001000000001    00000000        xxxxxxxx
# 5      0000    001000000001    00000000        zzzzzzzz
# 7      0001    001000000001    00000000        00000001
# 10     0001    000000000001    00000000        00000001
# 11     0001    000000000001    00000000        xxxxxxxx
# 13     0001    000000000001    00000000        zzzzzzzz
# 15     0010    000000000001    00000000        00000001
# 18     0010    100100000011    00000000        00000001
# 19     0010    100100000011    00000000        xxxxxxxx
# 21     0010    100100000011    00000000        zzzzzzzz
# 23     0011    100100000011    00000000        00000011
# 26     0011    011100000101    00000000        00000011
# 27     0011    011100000101    00000000        xxxxxxxx
# 29     0011    011100000101    00000000        zzzzzzzz
# 31     0100    011100000101    00000000        00000101
# 34     0100    100100000001    00000000        00000101
# 35     0100    100100000001    00000000        xxxxxxxx
# 37     0100    100100000001    00000000        zzzzzzzz
# 39     0101    100100000001    00000000        00000001
# 42     0101    100100000010    00000000        00000001
# 43     0101    100100000010    00000000        xxxxxxxx
# 45     0101    100100000010    00000000        zzzzzzzz
# 47     0110    100100000010    00000000        00000010
# 50     0110    101100000001    00000000        00000010
# 51     0110    101100000001    00000000        xxxxxxxx
# 53     0110    101100000001    00000000        zzzzzzzz
# 55     0111    101100000001    00000000        00000001
# 58     0111    010100000101    00000000        00000001
# 59     0111    010100000101    00000000        xxxxxxxx
# 61     0111    010100000101    00000000        zzzzzzzz
```

Cont.

```
# 63     1000    010100000101    00000000        00000101
# 66     1000    011000000000    00000000        00000101
# 67     1000    011000000000    00000000        xxxxxxxx
# 69     1000    011000000000    00000000        zzzzzzzz
# 71     1001    011000000000    00000000        00000000
# 74     1001    100100000001    00000000        00000000
# 75     1001    100100000001    00000000        xxxxxxxx
# 77     1001    100100000001    00000000        zzzzzzzz
# 79     1010    100100000001    00000000        00000001
# 82     1010    001100000111    00000000        00000001
# 83     1010    001100000111    00000000        xxxxxxxx
# 85     1010    001100000111    00000000        zzzzzzzz
# 87     1011    001100000111    00000000        00000111
# 90     1011    100100000010    00000000        00000111
# 91     1011    100100000010    00000000        xxxxxxxx
# 93     1011    100100000010    00000000        zzzzzzzz
# 95     1100    100100000010    00000000        00000010
# 98     1100    010000000111    00000000        00000010
# 99     1100    010000000111    00000000        xxxxxxxx
# 101    1100    010000000111    00000000        zzzzzzzz
# 103    1101    010000000111    00000000        00000111
# 106    1101    100000000000    00000000        00000111
# 107    1101    100000000000    00000000        xxxxxxxx
# 109    1101    100000000000    00000000        zzzzzzzz
# 111    1110    100000000000    00000000        00000000
```

Cont.

```
# RUN MODE
# 115    0000    100000000000    00000000        00000000
# 121    0000    100000000000    00000001        00000000
# 123    0001    100000000000    00000001        00000000
# 129    0010    100000000000    00000001        00000000
# 137    0010    100000000000    00000100        00000000
# 139    0011    100000000000    00000100        00000000
# 145    0101    100000000000    00000100        00000000
# 153    0101    100000000000    00000110        00000000
# 155    0110    100000000000    00000110        00000000
# 163    0110    100000000000    00000101        00000000
# 165    0111    100000000000    00000101        00000000
# 175    1000    100000000000    00000101        00000000
# 181    1010    100000000000    00000101        00000000
# 189    1011    100000000000    00000101        00000000
# 197    1011    100000000000    00000111        00000000
# 199    1100    100000000000    00000111        00000000
# 207    1100    100000000000    00000101        00000000
# 209    1101    100000000000    00000101        00000000
# 213    1101    100000000000    00000101        00000101
# 215    1110    100000000000    00000101        00000101
# 221    1111    100000000000    00000101        00000101
# ** Note: $finish    : C:/intelFPGA_lite/20.1/CPU/CPU_Ne
#    Time: 614 ps  Iteration: 0  Instance: /CU_TB_New
```

TB2 SIMULATION

WITH MOVW, ANDW, ORW, NOTW, MOVWM, ADDMW, SUBMW AND OUTPUTW

Testbench Module

```
module CU_TB;
        reg PROG_RUN, NEXT, CS, OE, WE, CLK;
        reg [11:0] INST;
        wire [7:0] PORT;
        wire [3:0] PC_CHECK;

        CU U1(PORT,PC_CHECK,INST,PROG_RUN,NEXT,CS,OE,WE,CLK);

        always #1 CLK = ~CLK;

        initial begin
                CLK = 0;
                PROG_RUN = 1; NEXT = 0;
                CS = 0; OE = 0; WE = 1;

                $display("CPU SIMULATION");
                $display("Time\tPC\tINST\t\tW\t\tPORT");
                $display("_____________________________________________");
                $monitor("%0t\t%b\t%b\t\t%b\t\t%b",
                        $time, U1.PC, INST, U1.W, PORT);


                $display("PROGRAM MODE");
```

Cont.

```
                $display("PROGRAM MODE");

                //MOVW ANDW ORW NOTW MOVWM SUBMW ADDMW OUTW

                //MOV 5 to W            PC 0000
                #2 INST = 12'b001000000101; #1 WE = 0; #1 WE = 1; #2 NEXT = 1; #2; NEXT = 0;

                //ANDW 255 to W         PC 0001
                #2 INST = 12'b110111111111; #1 WE = 0; #1 WE = 1; #2; NEXT = 1; #2; NEXT = 0;

                //ORW 5 to W            PC 0010
                #2 INST = 12'b111011111111; #1 WE = 0; #1 WE = 1; #2; NEXT = 1; #2; NEXT = 0;

                //NOTW 5 to W           PC 0011
                #2 INST = 12'b111100000000; #1 WE = 0; #1 WE = 1; #2; NEXT = 1; #2; NEXT = 0;

                //ADD 1 to W            PC 0100
                #2 INST = 12'b100100000001; #1 WE = 0; #1 WE = 1; #2; NEXT = 1; #2; NEXT = 0;

                //MOV W to M            PC 0101
                #2 INST = 12'b001100000111; #1 WE = 0; #1 WE = 1; #2 NEXT = 1; #2; NEXT = 0;

                //ADD 8 to W            PC 0110
                #2 INST = 12'b100100001000; #1 WE = 0; #1 WE = 1; #2; NEXT = 1; #2; NEXT = 0;

                //ADD M to W            PC 0111
                #2 INST = 12'b101000000111; #1 WE = 0; #1 WE = 1; #2; NEXT = 1; #2; NEXT = 0;

                //SUB M from W          PC 1000
                #2 INST = 12'b110000000111; #1 WE = 0; #1 WE = 1; #2; NEXT = 1; #2; NEXT = 0;

                //OUTPUT W              PC 1001
                #2 INST = 12'b100000000000; #1 WE = 0; #1 WE = 1; #2 NEXT = 1; #2; NEXT = 0;

                #2 PROG_RUN = 0;

                $display("RUN MODE");

                #500 $finish;
        end
endmodule
```

Transcript

```
# CPU SIMULATION
# Time   PC      INST            W               PORT
# ____________________________________________________________
# PROGRAM MODE
# 0      0000    xxxxxxxxxxxx    00000000        00000000
# 1      0000    xxxxxxxxxxxx    00000000        zzzzzzzz
# 2      0000    001000000101    00000000        zzzzzzzz
# 3      0000    001000000101    00000000        xxxxxxxx
# 5      0000    001000000101    00000000        zzzzzzzz
# 7      0001    001000000101    00000000        00000101
# 10     0001    110111111111    00000000        00000101
# 11     0001    110111111111    00000000        xxxxxxxx
# 13     0001    110111111111    00000000        zzzzzzzz
# 15     0010    110111111111    00000000        11111111
# 18     0010    111011111111    00000000        11111111
# 19     0010    111011111111    00000000        xxxxxxxx
# 21     0010    111011111111    00000000        zzzzzzzz
# 23     0011    111011111111    00000000        11111111
# 26     0011    111100000000    00000000        11111111
# 27     0011    111100000000    00000000        xxxxxxxx
# 29     0011    111100000000    00000000        zzzzzzzz
# 31     0100    111100000000    00000000        00000000
# 34     0100    100100000001    00000000        00000000
# 35     0100    100100000001    00000000        xxxxxxxx
# 37     0100    100100000001    00000000        zzzzzzzz
# 39     0101    100100000001    00000000        00000001
# 42     0101    001100000111    00000000        00000001
# 43     0101    001100000111    00000000        xxxxxxxx
# 45     0101    001100000111    00000000        zzzzzzzz
# 47     0110    001100000111    00000000        00000111
# 50     0110    100100001000    00000000        00000111
# 51     0110    100100001000    00000000        xxxxxxxx
# 53     0110    100100001000    00000000        zzzzzzzz
# 55     0111    100100001000    00000000        00001000
# 58     0111    101000000111    00000000        00001000
# 59     0111    101000000111    00000000        xxxxxxxx
# 61     0111    101000000111    00000000        zzzzzzzz
# 63     1000    101000000111    00000000        00000111
# 66     1000    110000000111    00000000        00000111
# 67     1000    110000000111    00000000        xxxxxxxx
# 69     1000    110000000111    00000000        zzzzzzzz
# 71     1001    110000000111    00000000        00000111
# 74     1001    100000000000    00000000        00000111
# 75     1001    100000000000    00000000        xxxxxxxx
# 77     1001    100000000000    00000000        zzzzzzzz
# 79     1010    100000000000    00000000        00000000
```

Cont.

```
# RUN MODE
# 83     0000    100000000000    00000000        00000000
# 89     0000    100000000000    00000101        00000000
# 91     0001    100000000000    00000101        00000000
# 101    0010    100000000000    00000101        00000000
# 109    0010    100000000000    11111111        00000000
# 111    0011    100000000000    11111111        00000000
# 119    0011    100000000000    00000000        00000000
# 121    0100    100000000000    00000000        00000000
# 129    0100    100000000000    00000001        00000000
# 131    0101    100000000000    00000001        00000000
# 139    0110    100000000000    00000001        00000000
# 147    0110    100000000000    00001001        00000000
# 149    0111    100000000000    00001001        00000000
# 159    0111    100000000000    00001010        00000000
# 161    1000    100000000000    00001010        00000000
# 171    1000    100000000000    00001001        00000000
# 173    1001    100000000000    00001001        00000000
# 177    1001    100000000000    00001001        00001001
# 179    1010    100000000000    00001001        00001001
# 185    1011    100000000000    00001001        00001001
# 191    1100    100000000000    00001001        00001001
# 197    1101    100000000000    00001001        00001001
# 203    1110    100000000000    00001001        00001001
# 209    1111    100000000000    00001001        00001001
# ** Note: $finish    : C:/intelFPGA_lite/20.1/CPU/CPU_New
#    Time: 582 ps  Iteration: 0  Instance: /CU_TB_New
```